

Global Climate Report — June 2026

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Summary of Notable Events, June 2026

- Global average temperature: second-warmest June on record.
- Arctic sea ice extent: third-lowest June on record. Global sea ice extent: fourth-lowest June on record.
- Arctic region: warmest June on record.
- North America: second-warmest June on record.
- Tropical Storm Arthur: weak, short-lived storm brought gusty winds and heavy rain to the Gulf Coast and southeastern U.S.
- Tropical Storm Cristina: tracked along the Central American coastline, produced prolonged heavy rainfall, triggered flash floods and river overflows in low-lying communities.
- Global tropical cyclone activity was above average, with seven named storms in June.
- Global ocean surface temperature departure: highest on record for the month.
- Europe: temperature departure tied as second-warmest June on record. One of the most severe early-summer heat waves on record, setting multiple temperature records.
- Asia: fourth-warmest June on record.
- India: one of its driest Junes on record.
- Super Typhoon Mekkhala: Category 4 equivalent typhoon; brought heavy rain, gusty winds, and storm surge to the Philippines, Taiwan, and Japan.
- Hong Kong: historic torrential rains caused widespread flooding and severe disruptions to daily life.
- Africa: second-warmest June on record.
- Australia: warmest June since 1966; fifth-warmest June on record.
- Oceania: temperature departure was the third-highest on record.
- New Zealand: warmest June on record.
- South America: warmer than average, but fell short of a top-10 ranking.
- Antarctic region: below-average June temperature. Sixth-lowest June sea ice extent on record.

Note: Material in the original report's map was compiled from NOAA's State of the Climate Reports. More information: <https://www.ncei.noaa.gov/access/monitoring/monthly-report/global>

Temperature

With the release of the January 2026 Global Climate Report, NOAA transitioned to version 6.1.0 of its Global Surface Temperature (NOAAGlobalTemp) dataset, replacing version 6.0.0. This introduces a new sea surface temperature dataset (ERSST v6). Specific anomalies and rankings may shift slightly, but short-term and long-term trends remain statistically consistent with version 6.0.0.

NOAA's National Centers for Environmental Information calculates the global temperature anomaly every month using NOAA GlobalTemp version 6.1.0, a reconstructed dataset covering both land and ocean surfaces. Because it is reconstructed, the entire period of record is recalculated each month with new data, so historical values can be updated over time. The most current reconstruction is always considered the most representative, and is publicly available through NOAA's Climate at a Glance tool.

June 2026 Global Temperature

The June 2026 global surface temperature was 1.09 degrees C (1.96 degrees F) above average — the second-warmest June in the 177-year record. This was 0.09 degrees C (0.16 degrees F) below the 2024 record, and only 0.02 degrees C (0.04 degrees F) above June 2023 (third-warmest June on record). The 10 warmest Junes on record have all occurred since 2015, with each of the last eight years (2019-2026) ranking among them. This was also the 50th consecutive June with global temperatures above the 20th-century average; the last below-average June occurred in 1976 (-0.05 degrees C / -0.09 degrees F).

The global ocean temperature departure for June was the highest on record at 0.99 degrees C (1.78 degrees F) above average, exceeding the previous June record set in 2024 by 0.04 degrees C (0.07 degrees F). Across all 2,118 months in the historical record since January 1850, June 2026 tied with July 2023 as the seventh-highest global ocean temperature departure on record. Global land surface temperatures ranked as the fourth-warmest on record for the month.

June featured warmer-than-average temperatures across much of the globe's land and ocean surfaces. Departures of +2.0 degrees C (+3.6 degrees F) or higher spanned Europe, eastern Australia, parts of Africa, the Arctic, the Weddell Sea, and a vast swath of Siberia extending into south-central Asia, as well as along the equator in the eastern and central tropical Pacific Ocean, where an active El Niño continues to strengthen. NOAA's Climate Prediction Center gives a 97% chance that this El Niño pattern will persist through early spring 2027. Record-high June temperatures were set across the eastern and central tropical Pacific Ocean, parts of Canada, northern South America, Africa, Europe, Asia, and parts of every major ocean basin.

Below-average temperatures covered much of Antarctica, resulting in its coolest June since 2021. Cooler-than-average pockets were also observed in parts of South America and Asia, and in northwestern Alaska and nearby Arctic waters. No land or ocean areas on Earth experienced record-cold temperatures for the month.

Regional Highlights

- Arctic: warmest June on record at 2.22 degrees C (4.00 degrees F) above average, beating the previous 2012 record by 0.26 degrees C (0.47 degrees F). First time the Arctic saw a June departure of 2.0 degrees C (3.6 degrees F) or more. The Arctic's 10 warmest Junes have all occurred since 2012.
- North America, Europe (tied), and Africa: each had their second-warmest June on record.
- Oceania: third-warmest June on record. Asia: fourth-warmest June on record.
- South America: above-average June temperature, but did not rank among the ten warmest Junes.

Europe Heat Wave

During mid- to late June, Europe was gripped by one of its most severe early-summer heat waves on record. Driven by a persistent high-pressure system, the heat wave pushed temperatures above 40

degrees C (104 degrees F) in Spain, France, and Italy. The extreme conditions strained power grids and heightened wildfire risks, particularly across the Mediterranean. It also severely disrupted transportation; in France and the United Kingdom, rail operators imposed speed restrictions and faced widespread delays because intense heat threatened to expand and buckle steel tracks. Record-breaking temperatures were set across the region.

Country Notes: Europe Heat Wave

- France: June heatwave surpassed the absolute intensity of the historic 2003 event, despite being two days shorter (14 days vs 16). From June 24-25, the national 24-hour mean temperature exceeded 30.0 degrees C (86.0 degrees F) for the first time on record, beating the previous all-time record of 29.4 degrees C (84.9 degrees F) set on August 5, 2003 and matched July 25, 2019. Meteo France simultaneously placed 72 departments under a red heatwave alert, the highest number since the alert system began in 2004. Over 40% of the nation exceeded 40.0 degrees C (104.0 degrees F) at least once during the heat wave. Nearly 75% of France endured 'tropical nights' with minimum temperatures above 20.0 degrees C (68.0 degrees F). Hundreds of local maximum and minimum records were broken.
- United Kingdom: shattered June temperature records on consecutive days in late June 2026. The record was first broken June 24 at 36.1 degrees C (97.0 degrees F), then surpassed the next day (June 25) when Yeovilton reached 36.4 degrees C (97.5 degrees F). Both values eclipsed the previous long-standing June record of 35.6 degrees C (96.1 degrees F), set in June 1957 and matched in June 1976. A Red Warning for extreme heat was issued for three consecutive days, a historic first for the country. The UK also provisionally broke its record for highest June minimum temperature when Bute Park, Cardiff, recorded an overnight low of 23.5 degrees C (74.3 degrees F) on June 25.
- Germany: 8-10 days had temperatures above 35.0 degrees C (95.0 degrees F) in parts of southwestern Germany, and temperatures exceeded 40.0 degrees C (104.0 degrees F) at 46 weather stations. A preliminary new all-time high of 41.7 degrees C (107.1 degrees F) was recorded on June 28 at Neisse-münde-Coschen, Brandenburg. Germany also recorded one of its warmest nights during the peak of the heat wave, at 29.4 degrees C (84.9 degrees F).
- Spain: at the peak of the heat wave, parts of Spain recorded temperatures above 40 degrees C (104 degrees F). On June 24, Bilbao Airport registered a maximum of 42.7 degrees C (108.9 degrees F), a new maximum temperature for June in the nation.
- The Netherlands: unprecedented extreme heat on June 26. In Ell, a new national June record was set at 39.4 degrees C (102.9 degrees F). The same day, Terschelling reached 33.6 degrees C (92.5 degrees F), the highest temperature ever recorded on the Dutch Wadden Island. Authorities issued a Code Red warning for extreme heat across much of the nation for the first time on record.

National June Temperature Summaries (preliminary, from national meteorological services and media reports)

- Netherlands: De Bilt recorded an average temperature of 19.1 degrees C (66.4 degrees F), 2.9 degrees C (5.2 degrees F) above average. Second-warmest June since records began in 1901, trailing the June 2023 record by 0.2 degrees C (0.4 degrees F).
- United Kingdom: second-warmest June on record, surpassed only by June 2023. England had its warmest June on record at 17.1 degrees C (62.8 degrees F), nearly 3.0 degrees C (5.4 degrees F)

above average, beating the previous record set in 2025. Wales had its second-warmest June; Scotland and Northern Ireland both tied for fourth-warmest since records began in 1884. A historic late-month heat wave produced the first consecutive three-day Red Extreme Heat Warning since that warning system was introduced in 2021, and drove record June average minimum temperatures for the UK, England, and Wales, while Scotland tied for highest.

- Spain: June temperature departure of +3.2 degrees C (+5.8 degrees F), the second-highest for June since national records began in 1961, trailing only the 2025 record. Per Spain's National Agency of Meteorology, 13 of the 15 warmest Junes have occurred in the 21st century.
- Switzerland: third-warmest June since national records began in 1864, with an average temperature 3.5 degrees C (6.3 degrees F) above average.
- France: warmest June on record with a national mean departure of 3.8 degrees C (6.8 degrees F) above the 1991-2020 period, surpassing the previous record set in June 2003 by 0.30 degrees C (0.54 degrees F).
- Germany: national average temperature of 19.5 degrees C (67.0 degrees F), the country's second-warmest June on record — 4.1 degrees C (7.4 degrees F) above the 1961-1990 average, and just 0.30 degrees C (0.54 degrees F) below the record-holding June of 2019.
- Austria: an exceptionally intense late-month heat wave shattered monthly temperature records at 157 of 266 weather stations. However, because temperatures during the first half of the month trended near or below normal, June ranked only as the fourth-warmest June on record for the nation.
- Japan: June temperature was 0.46 degrees C (0.83 degrees F) above the 1991-2020 average. It did not rank among the nation's five warmest Junes, and was Japan's lowest June temperature departure since 2019.
- Australia: exceptionally warm June, with a national mean temperature 1.50 degrees C (2.70 degrees F) above the 1961-1990 average — the highest June departure since 1966, and the fifth-highest since national records began in 1910. Warmth was most pronounced in the southeast: New South Wales had its second-warmest June; Victoria and Tasmania their third-warmest; South Australia its fifth-warmest; Queensland tied for seventh-warmest. New South Wales and Victoria recorded their second-warmest June minimum temperatures, and South Australia its fourth-warmest. Nationally, average minimum temperature ranked fifth-warmest on record; average maximum temperature ranked ninth-highest.
- New Zealand: warmest June since national records began in 1909, at 10.6 degrees C (51.1 degrees F), 1.9 degrees C (3.4 degrees F) above average. On June 20, Bromley (Christchurch) reached 26.0 degrees C (78.8 degrees F), New Zealand's highest June temperature on record.

June 2026 Ranks and Records (selected, out of 177-year record: 1850-2026)

Format: Region — Departure from 1901-2000 average — Rank among warmest Junes.

- Global Land: +1.34 degrees C — 4th warmest
- Global Ocean: +0.99 degrees C — 1st warmest (record)
- Global Land and Ocean: +1.09 degrees C — 2nd warmest
- Northern Hemisphere Land: +1.75 degrees C — 2nd warmest

- Northern Hemisphere Ocean: +1.16 degrees C — 2nd warmest
- Northern Hemisphere Land and Ocean: +1.42 degrees C — 2nd warmest
- Southern Hemisphere Land: +0.39 degrees C — 39th warmest
- Southern Hemisphere Ocean: +0.86 degrees C — 1st warmest (record)
- Southern Hemisphere Land and Ocean: +0.77 degrees C — 3rd warmest
- North America: +1.70 degrees C — 2nd warmest
- South America: +0.89 degrees C — 19th warmest
- Europe: +2.38 degrees C — 2nd warmest (tied with 2024)
- Africa: +1.54 degrees C — 2nd warmest
- Asia: +1.79 degrees C — 4th warmest
- Oceania: +1.37 degrees C — 3rd warmest
- Arctic: +2.22 degrees C — 1st warmest (record)
- Antarctic: -0.45 degrees C — 151st warmest (i.e., cool relative to record; tied with 1936 in coolest ranking context)
- Caribbean Islands: +0.88 degrees C — 8th warmest
- Atlantic Main Development Region (MDR): +0.66 degrees C — 17th warmest
- Gulf of America: +0.95 degrees C — 9th warmest
- Hawaiian Region: +1.06 degrees C — 3rd warmest
- East North Pacific: +1.76 degrees C — 1st warmest (record)

Year-to-Date Temperature: January-June 2026

The January-June global surface temperature departure was the third-highest in the 177-year record at +1.14 degrees C (+2.05 degrees F), surpassed only by the corresponding periods in 2024 and 2025. The 10 warmest January-June periods have all occurred since 2016, with each of the last five years (2022-2026) ranking among them. NCEI's statistical analysis indicates it is virtually certain that 2026 will rank among the 10 warmest years on record, and very likely it will place within the top five.

Global oceans continued to reach near-record levels, with a January-June temperature departure of +0.91 degrees C (+1.64 degrees F). The global land surface temperature ranked as the fifth-highest on record for the January-June period.

The first half of 2026 was characterized by widespread, warmer-than-average conditions globally. Departures of 2.0 degrees C (3.6 degrees F) or higher were observed across the contiguous U.S., the Arctic, parts of Asia, and the North Pacific Ocean. Record-high January-June temperatures occurred in parts of the contiguous U.S., Mexico, Europe, Asia, Africa, northern Chile, and across every major ocean basin. Below-average temperatures were limited to parts of Alaska into northern Canada, northern Asia, and northeastern Europe. No land or ocean areas experienced record-cold temperatures during this six-month period.

Several continents and regions ranked among their top-10 warmest January-June periods on record: Africa (3rd-warmest), Asia (5th-warmest), North America (8th-warmest), and South America and Oceania (each 10th-warmest). Europe, the Arctic, and Antarctic also had warmer-than-average conditions but did

not rank in their respective top 10.

January-June 2026 Ranks and Records (selected)

- Global Land: +1.64 degrees C — 5th warmest
- Global Ocean: +0.91 degrees C — 2nd warmest
- Global Land and Ocean: +1.14 degrees C — 3rd warmest
- Northern Hemisphere Land: +1.98 degrees C — 5th warmest
- Northern Hemisphere Ocean: +0.99 degrees C — 2nd warmest
- Northern Hemisphere Land and Ocean: +1.42 degrees C — 5th warmest
- Southern Hemisphere Land: +0.88 degrees C — 6th warmest
- Southern Hemisphere Ocean: +0.85 degrees C — 2nd warmest
- Southern Hemisphere Land and Ocean: +0.86 degrees C — 2nd warmest
- North America: +1.65 degrees C — 8th warmest
- South America: +1.08 degrees C — 10th warmest
- Europe: +1.54 degrees C — 12th warmest
- Africa: +1.46 degrees C — 3rd warmest
- Asia: +2.19 degrees C — 5th warmest
- Oceania: +0.98 degrees C — 10th warmest
- Arctic: +2.11 degrees C — 12th warmest
- Antarctic: +0.39 degrees C — 11th warmest
- Caribbean Islands: +0.89 degrees C — 6th warmest
- Atlantic MDR: +0.73 degrees C — 12th warmest
- Gulf of America: +0.94 degrees C — 6th warmest
- Hawaiian Region: +1.07 degrees C — 1st warmest (record)
- East North Pacific: +1.41 degrees C — 1st warmest (record)

Precipitation

Precipitation data from the Global Historical Climatology Network (GHCN) are with respect to the 1961-1990 average.

During June 2026, global precipitation patterns showed stark regional contrasts. Above-average precipitation was prominent across the central and eastern U.S., northern Argentina, northern Europe, northwestern and eastern parts of Asia, and southern and southeastern Australia. Precipitation deficits were recorded across southern and western Alaska, the western and southwestern contiguous U.S., western, central, and eastern Europe, parts of western and eastern Australia, much of Argentina and Uruguay, and a vast portion of northern and eastern Siberia.

Notable Precipitation Events (preliminary)

- Southeastern Europe: a severe thunderstorm outbreak swept across the region June 10-12, 2026, bringing destructive winds, torrential rainfall, and golf-ball-sized hail to parts of Slovenia, Croatia,

Bosnia and Herzegovina, and Greece, damaging more than 100 buildings.

- Spain: third-driest June since national records began in 1961, receiving just 39% of normal precipitation.
- Hong Kong, China: multiple episodes of torrential rain during June 2026; authorities issued three Black Rainstorm Warnings (the territory's highest rainfall alert level), including a historic event on June 18 when two Black Rainstorm Warnings were issued within 3.5 hours of each other — the first such occurrence since the warning system began in 1992. Rainfall rates exceeded 70 mm per hour in some areas. The New Territories were hit particularly hard, with roads, border facilities, and low-lying districts flooded; schools closed, flights disrupted, and landslide/flash flood warnings issued.
- Tropical Storm Cristina: formed off the Pacific coast of Nicaragua on June 8, 2026. Tracked slowly along the Central American coastline, producing prolonged heavy rainfall. Most regions received 100-200 mm (4-8 inches) of rain, with isolated areas in Central America reporting as much as 300 mm (12 inches). Triggered flash flooding, river overflows, and landslides across Nicaragua, El Salvador, and Guatemala. In El Salvador alone, nearly 200 homes were reported damaged or destroyed.
- Italy: severe thunderstorms brought localized flash flooding, strong winds, and hail to parts of Italy in late June, particularly northern regions, causing travel disruptions and localized damage.
- Australia: wettest June since 2016, with a nationwide rainfall total 27% above average. Every state was wetter than normal; South Australia had the largest departure at 130% above average. Driven by cold fronts early and late in the month, plus a mid-month northwest cloudband. Triggered minor flooding in parts of Victoria and Tasmania and broke numerous June daily rainfall records.
- India: unusually dry, one of its driest Junes since 1901. National rainfall total of 99.5 mm (3.9 inches), just 60% of normal precipitation, primarily due to a weak monsoon and its delayed advancement over northwestern India.

Regional Background

The regional analysis, introduced in July 2015, focuses on monitoring temperature departures across individual continents and key regions. Continental and regional temperature departures are derived from the NOAA GlobalTemp dataset. Continental and regional polygons are mapped using 5-degree x 5-degree grid points centered within each area; these grid points are area-weighted and averaged to compute a localized temperature departure time series. This complements global monitoring by showing how the last month, season, or year compares to the past for each continent and key region.

Named sub-regions referenced in this report include: Caribbean Islands, Hawaiian Region, East North Pacific, Gulf of America, and the Atlantic Main Development Region (Atlantic MDR) — all defined as specific geographic boxes near North America, Central America, and the tropical Atlantic/Pacific, as shown in the original report's reference maps.

References

- Adler, R., G. Gu, M. Sapiiano, J. Wang, G. Huffman (2017). Global Precipitation: Means, Variations and Trends During the Satellite Era (1979-2014). *Surveys in Geophysics* 38: 679-699.
doi:10.1007/s10712-017-9416-4

- Adler, R., M. Sapiano, G. Huffman, J. Wang, G. Gu, D. Bolvin, L. Chiu, U. Schneider, A. Becker, E. Nelkin, P. Xie, R. Ferraro, D. Shin (2018). The Global Precipitation Climatology Project (GPCP) Monthly Analysis (New Version 2.3) and a Review of 2017 Global Precipitation. *Atmosphere*. 9(4), 138. doi:10.3390/atmos9040138
- Gu, G., and R. Adler (2022). Observed Variability and Trends in Global Precipitation During 1979-2020. *Climate Dynamics*. doi:10.1007/s00382-022-06567-9
- Huang, B., Peter W. Thorne, et al. (2017). Extended Reconstructed Sea Surface Temperature version 5 (ERSSTv5), Upgrades, validations, and intercomparisons. *J. Climate*. doi:10.1175/JCLI-D-16-0836.1
- Huang, B., V.F. Banzon, E. Freeman, J. Lawrimore, W. Liu, T.C. Peterson, T.M. Smith, P.W. Thorne, S.D. Woodruff, and H-M. Zhang (2016). Extended Reconstructed Sea Surface Temperature Version 4 (ERSST.v4). Part I: Upgrades and Intercomparisons. *J. Climate*, 28, 911-930. doi:10.1175/JCLI-D-14-00006.1
- Menne, M.J., C.N. Williams, B.E. Gleason, J.J. Rennie, and J.H. Lawrimore (2018). The Global Historical Climatology Network Monthly Temperature Dataset, Version 4. *J. Climate*, in press. <https://doi.org/10.1175/JCLI-D-18-0094.1>
- Peterson, T.C. and R.S. Vose (1997). An Overview of the Global Historical Climatology Network Database. *Bull. Amer. Meteorol. Soc.*, 78, 2837-2849.
- Vose, R., B. Huang, X. Yin, D. Arndt, D.R. Easterling, J.H. Lawrimore, M.J. Menne, A. Sanchez-Lugo, and H.M. Zhang (2021). Implementing Full Spatial Coverage in NOAA's Global Temperature Analysis. *Geophysical Research Letters* 48(10), e2020GL090873. doi:10.1029/2020gl090873